

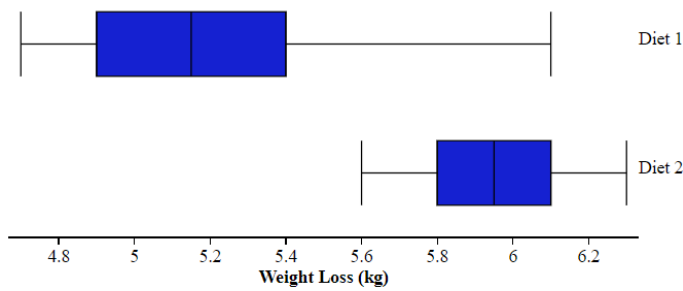
HW 4: Confidence Intervals for a Difference in Means

1) Do high school girls and boys differ in how many hours of sleep they get per night? At your high school, 35 girls were randomly selected and 40 boys were randomly selected. They were asked how many hours of sleep they got last night. Their answers were recorded and the summary statistics are listed below:

	n	$\bar{x}$	s
Girls	35	6.55 hours	0.45 hours
Boys	40	6.21 hours	0.22 hours

Construct and interpret a 90% confidence interval for a difference in Girls – Boys in the amount of sleep they get per night.

2) Suppose a researcher is studying the effect of two different diets on weight loss. The researcher randomly assigns 10 participants to Diet 1 and another 10 participants to Diet 2. After 8 weeks, the weight loss (in kilograms) for each participant is recorded. Here are the summary statistics for this study.



Group	n	Sample Mean	Sample Std. Dev.
Diet 1	10	5.2	0.408
Diet 2	10	5.95	0.217

95% Confidence Interval Results:

μ_1 : mean weight loss of Diet 1

μ_2 : mean weight loss of Diet 2

$\mu_1 - \mu_2$: mean difference (no pooled variances)

Difference	DF	Lower Limit	Upper Limit
$\mu_1 - \mu_2$	13.721	-1.064	-0.436

a) The endpoints of this confidence interval are negative. What does this indicate?

b) What does the fact that the confidence interval not containing 0 indicate?

c) What is the margin of error for this confidence interval?

d) If we had created a 98% confidence interval instead, would the margin of error be larger or smaller? Explain.

e) Interpret the confidence interval in the context of the problem.